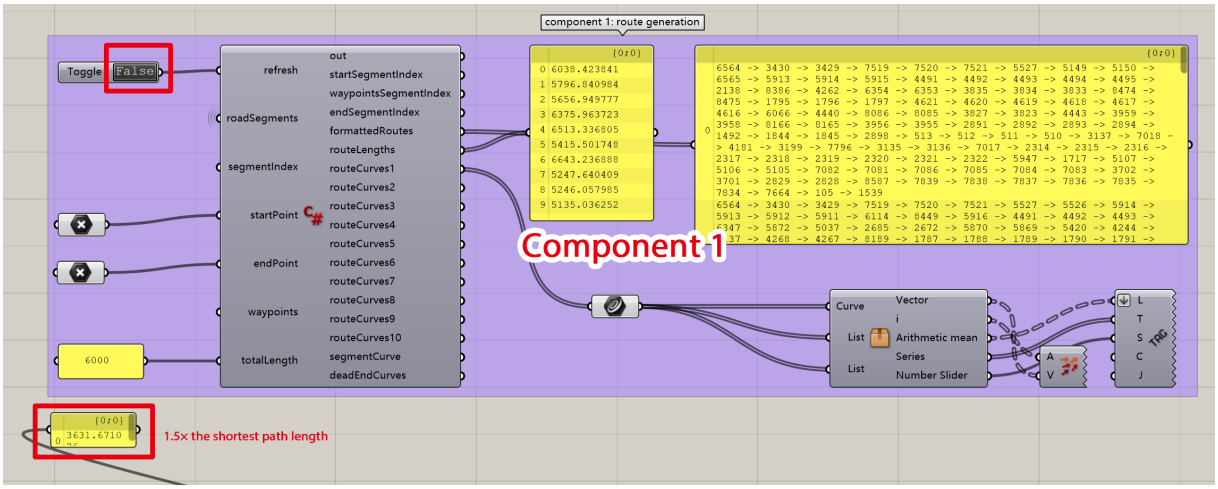
How to Run the Model

1. **Open the files**
   Open **mapping.3dm** in Rhino 7. Then open **algorithm.gh** in Grasshopper within Rhino.

2. **Generate candidate paths**
   Double-click the **"False" toggle** in **Component 1** to switch it to "True".
   The system will start generating 10 candidate paths.

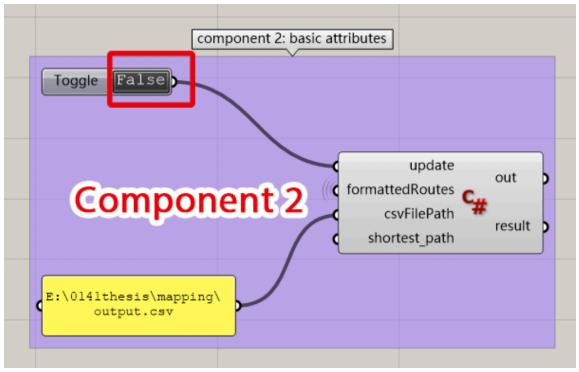
   - Please allow a few minutes for the computation to complete.
   - Do not perform other operations during this process.
   - You can monitor the progress through the **Panel** in Rhino, which displays textual updates of the generation process.



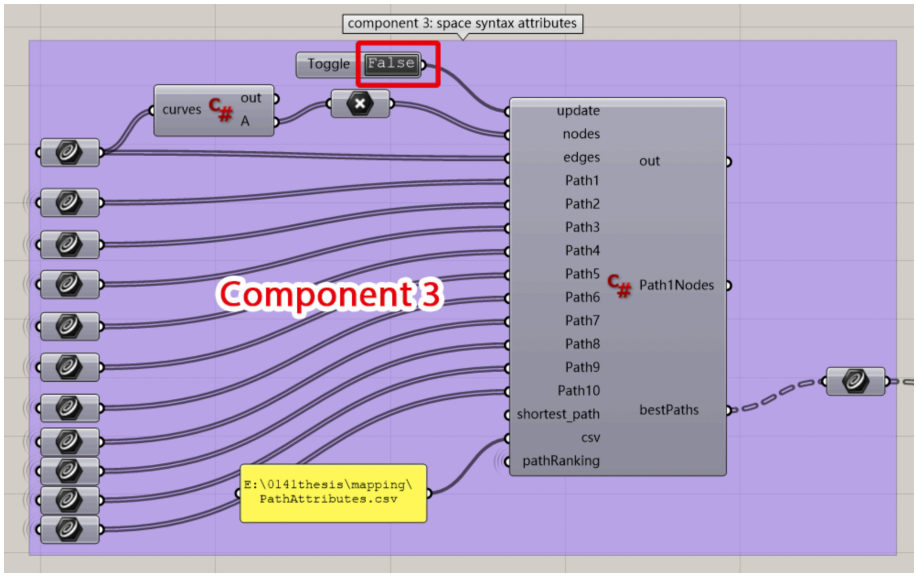
Notes:

- The code is set to stop after 300,000 seconds by default. If 10 paths are not successfully generated within this time, you may repeat the process.
- You can adjust:
  - startPoint
  - endPoint
  - add waypoints
  - target path length
- The target path length should be at least **1.5× the shortest path length** (shown in the Panel), otherwise path generation may fail.

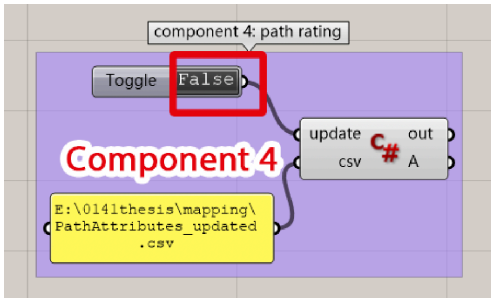
3. **Extract path attributes**
   Double-click the **"False" toggle** in **Component 2** to switch it to "True" to generate the file **PathAttributes.csv**
   (Make sure to set the correct file path before running.)



4. **Update path attributes**
   Double-click the **"False" toggle** in **Component 3** to switch it to "True" to generate **PathAttributes_updated.csv**



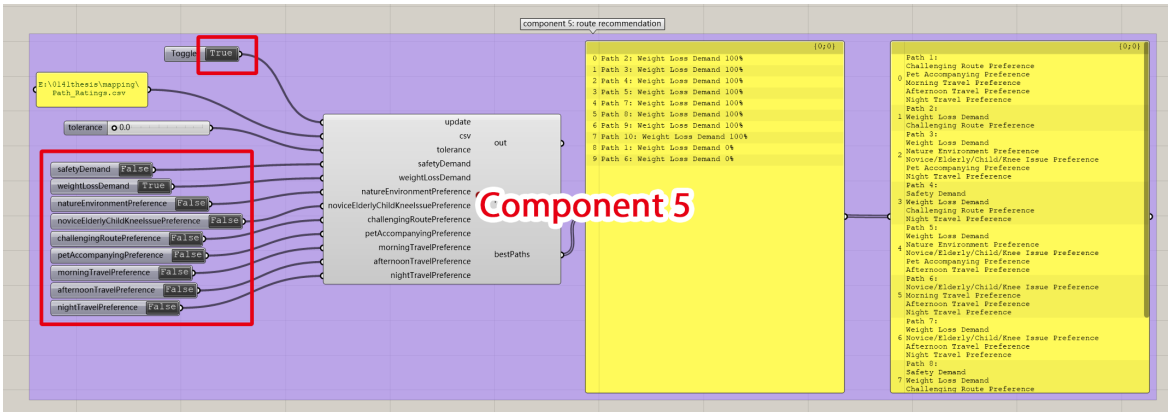
5. **Calculate path ratings**
   Double-click the **"False" toggle** in **Component 4** to switch it to "True" to generate **Path_Ratings.csv**

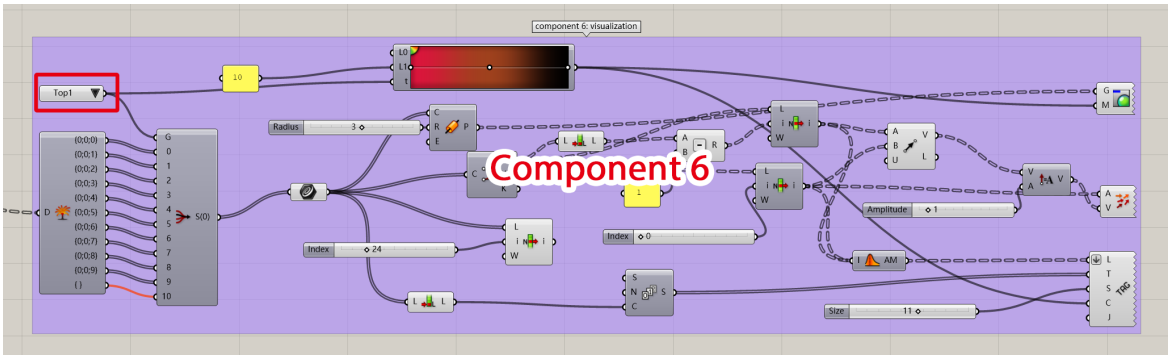


6. **Apply user preferences and ranking**

Double-click the **"False" toggle** in **Component 5** to switch it to "True" and select the desired **preference options**.

The system will output ranked path results based on the selected preferences.



7. **Visualise top paths**

Adjust the **topX button** in **Component 6** to visualise the selected top-ranked paths.